Date: 29/05/2026

NATIONAL SKILL TRAINING INSTITUTE (WOMEN), INDORE

ARTIFICIAL INTELLIGENCE PROGRAMMING ASSISTANT
(MODULE 9 - INTERNAL PRACTICAL)
2025-26

Set B

TIME: 3 Hrs **MARKS: 50**

Note: Attempt any 2

1. To understand how computers read and understand human language using pre-trained AI models, and to analyze a simple user review step by step.

Problem Statement
You are given a short customer review like:

"The product is very good and easy to use."

The task:

1. Load a Pre-trained Model
    a. A pre-trained model is an AI model that already knows a lot about language.
    b. You will load a simple model (like BERT) that can understand text.

    Think of it as loading a “brain” that already knows English.

2. Load the Tokenizer
    a. A tokenizer breaks sentences into smaller parts (words or pieces of words).
    b. This helps the model understand the sentence better.

    Example: "The product is good" → ["The", "product", "is", "good"]

3. Tokenize the Input Text
    a. Convert the review sentence into tokens.
    b. Also convert words into numbers (because computers understand numbers, not words).
4. Inspect Tokenization Output
    a. Inspect Tokenization Output
    b. Token IDs (numbers)

    This helps you see how the computer “reads” the sentence.

5. Run a Forward Pass Through the Model
    a. Pass the tokenized input into the model.

b. The model will process the sentence and give output (like understanding or prediction).

   Think of this as the AI “thinking” about the sentence.

2. A company wants to build a simple AI writing assistant that can automatically generate text from a given sentence or topic. The system should use Generative AI to complete sentences, generate short paragraphs, and create simple text outputs using Hugging Face Transformers.

   Task:

   - Load Required Libraries and Model
   - Input a Starting Sentence
   - Generate Text Automatically
   - Generate Text for Multiple Inputs
   - Display and Interpret the Results

   Note:

   - Use Python programming language.
   - Use the transformers library for text generation.
   - Install required libraries before running the program.
   - Use simple input sentences for testing.
   - Display generated text clearly.
   - Write comments for each task in the code.

3. Canva’s AI Image Generator enables users to create high-quality visuals from text prompts, making design faster and more accessible. In this lab, you will explore how generative AI can be used to create, refine, and apply visuals for real-world use cases such as education, marketing, and presentations.

   Task:

   1. Access Canva’s AI Image Generator
   2. Write an Effective Text Prompt
   3. Generate and Refine Images
   4. Customize and Edit the Visual
   5. Export and Evaluate the Output

   Task 1: Access Canva’s AI Image Generator

   Log in to Canva and navigate to Apps → AI Image Generator (Magic Media → Text to Image).

Task 2: Write an Effective Text Prompt

Create a detailed prompt describing the visual you want to generate.

> “A modern AI classroom with students using laptops, futuristic design, bright colors, flat illustration style”

Task 3: Generate and Refine Images

Generate multiple image variations and refine results by:

- Changing styles (illustration, photo, 3D, watercolor)
- Adjusting mood and color tone
- Modifying prompt keywords

Task 4: Customize and Edit the Visual

Select the best-generated image and enhance it using Canva tools:

- Add text, icons, and shapes
- Resize for presentation, poster, or social media
- Apply filters and layout adjustments

Task 5: Export and Evaluate the Output

- Download the final visual and evaluate:
- Prompt relevance
- Visual clarity and creativity
- Suitability for the intended use case